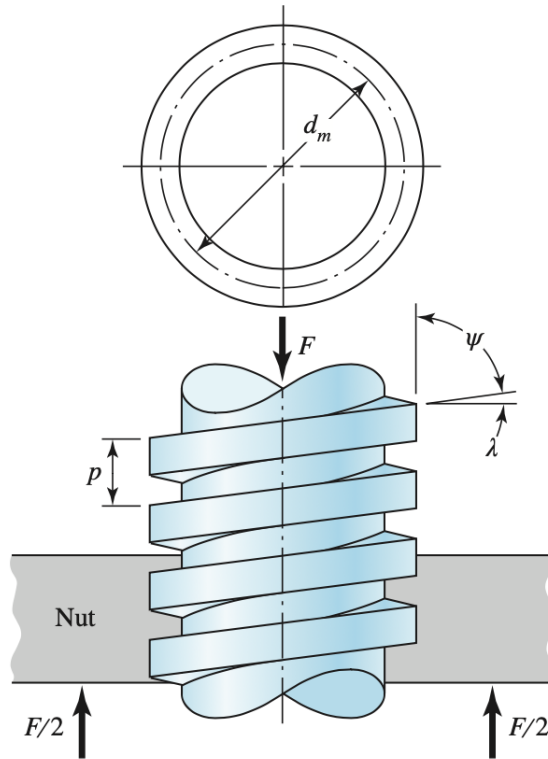


5 Springies, Bolties, Screwies (Boing Boing)

5.1 Power Screws

5.1.1 Anatomy



5.1.2 Design Selection

1. Calculate the required tensile stress area A_t in in^2 :

$$A_t = \frac{F}{\sigma_d}$$

where F is the load applied to the screw in lbs,

σ_d is maximum tensile stress in psi

2. Select a screw from one of the following tables (based on your screw type) that would provide a tensile stress area A_t greater or equal to the one calculated above.
Take note of the rated A_s .

TABLE 17-1 Preferred Acme Screw Threads

Nominal major diameter, D (in)	Threads per in, n	Pitch, $p = 1/n$ (in)	Minimum minor diameter, D_r (in)	Minimum pitch diameter, D_p (in)	Tensile stress area, A_t (in ²)	Shear stress area, A_s (in ²) ^a
1/4	16	0.0625	0.1618	0.2043	0.026 32	0.3355
5/16	14	0.0714	0.2140	0.2614	0.044 38	0.4344
3/8	12	0.0833	0.2632	0.3161	0.065 89	0.5276
7/16	12	0.0833	0.3253	0.3783	0.097 20	0.6396
1/2	10	0.1000	0.3594	0.4306	0.1225	0.7278
5/8	8	0.1250	0.4570	0.5408	0.1955	0.9180
3/4	6	0.1667	0.5371	0.6424	0.2732	1.084
7/8	6	0.1667	0.6615	0.7663	0.4003	1.313
1	5	0.2000	0.7509	0.8726	0.5175	1.493
1 $\frac{1}{8}$	5	0.2000	0.8753	0.9967	0.6881	1.722
1 $\frac{1}{4}$	5	0.2000	0.9998	1.1210	0.8831	1.952
1 $\frac{3}{8}$	4	0.2500	1.0719	1.2188	1.030	2.110
1 $\frac{1}{2}$	4	0.2500	1.1965	1.3429	1.266	2.341
1 $\frac{3}{4}$	4	0.2500	1.4456	1.5916	1.811	2.803
2	4	0.2500	1.6948	1.8402	2.454	3.262
2 $\frac{1}{4}$	3	0.3333	1.8572	2.0450	2.982	3.610
2 $\frac{1}{2}$	3	0.3333	2.1065	2.2939	3.802	4.075
2 $\frac{3}{4}$	3	0.3333	2.3558	2.5427	4.711	4.538
3	2	0.5000	2.4326	2.7044	5.181	4.757
3 $\frac{1}{2}$	2	0.5000	2.9314	3.2026	7.388	5.700
4	2	0.5000	3.4302	3.7008	9.985	6.640
4 $\frac{1}{2}$	2	0.5000	3.9291	4.1991	12.972	7.577
5	2	0.5000	4.4281	4.6973	16.351	8.511

^aPer inch of length of engagement.

TABLE 17-1M Examples of Power Screws with Metric Trapezoidal Screw Thread

ISO thread system—External threads

Major diameter, D (mm)	Pitch, p (mm)	Pitch diameter, D_p (mm)	Minor diameter, D_r (mm)	Tensile stress area (mm ²)
8	1.5	7.25	6.2	35.52
10	2	9.0	7.5	53.46
12	3	10.5	8.5	70.88
14	3	12.5	10.5	103.9
16	3	14.5	12.5	143.1
20	4	18.0	15.5	220.4
22	5	19.5	16.5	254.5
24	5	21.5	18.5	314.2
28	5	25.5	22.5	452.4
30	6	27.0	23.0	490.9
32	6	29.0	33.0	754.8
36	6	33.0	29.0	754.8
40	7	36.5	32.0	921.3
46	8	42.0	37.0	1225
50	8	46.0	41.0	1486
55	9	50.5	45.0	1791
60	9	55.5	50.0	2185
70	10	65.0	59.0	3019
80	10	75.0	69.0	4072
90	12	84.0	77.0	5090
100	12	94.0	87.0	6433
120	14	113.0	104.0	9246
125	14	122.0	109.0	10 477

3. Calculate the required length h of the nut/yoke in inches to maintain shear stress below the maximum shear stress.

- (a) Calculate the required shear area A_s in in² and call it:

$$A_s = \frac{F}{\tau_d}$$

where F is the load applied to the screw in lbs,

τ_d is the maximum shear stress in psi

(b) the required length h is:

$$h = A_{sc} \frac{1 \text{ in}}{A_{sr}}$$

where A_{sc} is the calculate shear area,

A_{sr} is the rated shear area obtained from the table

- Round up to the nearest 1/4th of an inch. For example, if you obtain 1.10 in, round up to 1.25 in. If you obtain 1.58, round up to 1.75 in, etc.

4. Compute the lead angle λ using the following equation:

$$\lambda = \tan^{-1} \frac{p}{\pi D_p}$$

where p is the pitch (obtained from the table) in inches,

D_p is the pitch diameter (obtained from the table) in inches

5. Determine the raising and lowering torque (T_u and T_d respectively, in lb · in).

(a) The equation to compute the raising torque is:

$$T_u = \frac{FD_p}{2} \left[\frac{\cos \phi \tan \lambda + f}{\cos \phi - f \tan \lambda} \right]$$

where λ is the lead angle,

f is the friction coefficient,

F is the load applied to the screw in lbs,

D_p is the pitch diameter (obtained from the table) in inches,

ϕ is the thread angle

- Square threads: $\phi = 0^\circ$
- Acme threads: $\phi = 14.5^\circ$
- Trapezoidal threads: $\phi = 15^\circ$

(b) The equation to compute the lowering torque is:

$$T_d = \frac{FD_p}{2} \left[\frac{f - \cos \phi \tan \lambda}{\cos \phi + f \tan \lambda} \right]$$

(c) If a collar friction is given, then the formulas for T_u and T_d are changed:

$$T_u = \frac{FD_p}{2} \left[\frac{\cos \phi \tan \lambda + f}{\cos \phi - f \tan \lambda} \right] + f_c F R_c$$

$$T_d = \frac{FD_p}{2} \left[\frac{f - \cos \phi \tan \lambda}{\cos \phi + f \tan \lambda} \right] + f_c F R_c$$

where f_c is the collar friction,

R_c is the collar friction radius

Note that if there is a roller bearing then you can assume that the collar friction is zero (unless stated otherwise).

6. Compute the efficiency e of the power screw:

$$e = \frac{Fp}{2\pi T_u}$$

where F is the load applied to the screw in lbs,

p is the pitch in inches,

T_u is the raising torque in lb · in

7. Calculate the maximum speed you can raise a load given an input power in hp.

- (a) Find the rotational speed of the screw (in rpm) using:

$$n = \frac{63000P}{T}$$

where P is the input power in hp,

T is the raising torque in lb · in

- (b) Convert to linear speed (in in/s) using the following relation:

$$V = \left(\frac{n \text{ rev}}{\text{min}} \right) \left(\frac{p \text{ in}}{\text{rev}} \right) \left(\frac{\text{min}}{60 \text{ sec}} \right)$$

where n is the rotational speed in rpm,

p is the pitch in in

5.2 Ball Screws

5.2.1 Design Selection

1. Compute the travel life in inches:

$$\text{Travel Life} = \left(\frac{\text{distance (in)}}{\text{cycle}} \right) \left(\frac{\# \text{ of cycles}}{\text{hour}} \right) \left(\frac{24 \text{ hours}}{\text{day}} \right) \left(\frac{365 \text{ days}}{\text{year}} \right) (\# \text{ of years})$$

2. Select a ball screw based on load and travel life: